Student Patent Novelty Check

Provide more detail → better retrieval. Use keywords and filters to steer the search.

LIVE · patentsview_live_semantic_anchor_gate (strict_anchors+keywords)

Backend health

Search summary

Copy this into your notes or for debugging.

Copy

Mode	patentsview_live_semantic_anchor_gate (strict_anchors+keywords)
Problem	Real-time fraud detection for card transactions by identifying coordinated fraud rings across merchants, devices, and accounts.
Novelty	Uses graph neural networks to model relationships between cardholder–merchant–device–transaction and detect collusive fr…
Technologies	AI/ML, Cloud, Database, Networking
Keywords	fraud, transaction, chargeback, authorization, merchant, cardholder, device fingerprint, graph neural network, anomaly detection
Exclude	inkjet, printer, nozzle, tire, vehicle, engine, reservoir, crop, drone, medical imaging
Assignee	Visa
Years	2015 → 2024

Describe your idea

More input usually improves relevance — especially keywords + exclusions.

What problem are you solving?

Real-time fraud detection for card transactions by identifying coordinated fraud rings across merchants, devices, and accounts.

What does your idea do?

Automates process

Analyzes data

- [ ] Reduces time
- [ ] Improves UX
- [ ] Hardware control

Primary domain

Software

Tip: Domain doesn't change semantic similarity much; keywords help more.

Key technologies

- [ ] AI/ML
- [ ] Cloud
- [ ] Mobile
- [ ] IoT
- [ ] Networking
- [ ] NLP
- [ ] Database

What's novel about your idea? (optional)

Uses graph neural networks to model relationships between cardholder–merchant–device–transacti

Power inputs

Advanced

These steer the search hard. Comma or new-line separated.

Must-include keywords

fraud, transaction, chargeback, authorization, merchant, cardholder, device fingerprint, graph neural network, anomaly detection

Exclude keywords

inkjet, printer, nozzle, tire, vehicle, engine, reservoir, crop, drone, medical imaging

Assignee (optional)

Visa

Max results

10

Year from

2015

Year to

2024

Do you want results to show a short abstract preview?

Note: This will only display if the backend returns `abstract` or `abstract_snippet` in results.

Find Similar Patents

Results

Idea about: Real-time fraud detection for card transactions by identifying coordinated fraud rings across merchants, devices, and accounts. · Domain: Software · Mode: patentsview_live_semantic_anchor_gate (strict_anchors+keywords)

CPC used

G06F

G06N

LIVE · patentsview_live_semantic_anchor_gate (strict_anchors+keywords)

10 results

2 CPC

Top match: 60%

Sort Relevance

Min similarity 0%

Reset

#1

Fraud reduction system for transactions

US10089683 · 2018 · Visa International Service Association

G06Q

Medium confidence

Similarity

60% Top match

Show abstract preview

A system, apparatus, and method for reducing fraud in payment or other transactions by providing issuers with a warning that a transaction being processed for authorization is potentially fraudulent. In some embodiments,…

Hide why similar

Also automates part of the workflow you described.

Uses data analysis to reach decisions, similar to your idea.

Falls under a similar domain (Software) in the CPC scheme.

Mentions technologies that overlap with yours: AI/ML, Cloud, Database, Networking

Classified in CPC area G06Q, which is related to your idea's focus.

Feedback

Helps us learn what “relevant” means for students.

👍 Relevant 👎 Not relevant

Optional: what did you expect to see instead?

Submit feedback

#2

Enhanced merchant identification using transaction data

Similarity **49%**

US10304056 · 2019 · Visa International Service Association

G06Q Low confidence

Show abstract preview

A system for identifying and associating merchant attributes data of a merchant with transaction data may be integrated with existing payment processing systems to provide for detailed merchant attributes data (e.g., mer…

Show why similar

Feedback

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👍 Relevant 👎 Not relevant

Optional: what did you expect to see instead?

Submit feedback

#3

Systems and methods to detect changes in merchant identification information

Similarity **46%**

US10269077 · 2019 · Visa International Service Association

G06Q Low confidence

Show abstract preview

A system and method to detect changes in transaction parameters for the identification of a store of the merchant in a payment processing network. The system is configured to monitor input data, such as acquirer's mercha…

Show why similar

Feedback

Helps us learn what “relevant” means for students.

👍 **Relevant** 👎 **Not relevant**

Optional: what did you expect to see instead?

Submit feedback

#4

Processor issuer detection and user level stand-in authorization

Similarity

45%

US10282709 · 2019 · Visa International Service Association

G06Q **Low confidence**

Show abstract preview

Embodiments of the invention are directed to systems and methods for stand-in processing using data driven capabilities. Transaction activity of an issuer may be monitored to detect a major incident and to automatically…

Show why similar

Feedback

Helps us learn what “relevant” means for students.

👍 **Relevant** 👎 **Not relevant**

Optional: what did you expect to see instead?

Submit feedback

#5

Method and system for correlating diverse transaction data

Similarity
42%

US10296904 · 2019 · Visa International Service Association

G06Q Low confidence

Show abstract preview

Embodiments of the present invention are directed to methods, systems, and apparatuses for matching e-commerce and physical point-of-sale transaction data. In one embodiment, a method includes receiving, by a computer, f…

Show why similar

Feedback

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👍 Relevant 👎 Not relevant

Optional: what did you expect to see instead?

Submit feedback

#6

Multi-device transaction verification

Similarity
41%

US10210521 · 2019 · Visa International Service Association

G06Q Low confidence

Show abstract preview

When a user enters a resource provider location with a portable communication device, the portable communication device provides an indication to a transaction processing system that the portable communication device is…

Show why similar

Feedback

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👍 Relevant 👎 Not relevant

Optional: what did you expect to see instead?

Submit feedback

#7

System and method for authorizing financial transactions with online merchants

Similarity
40%

US10008067 · 2018 · Visa U.S.A. Inc.

G07F Low confidence

Show abstract preview

A system for authorizing a financial transaction of a financial presentation device such as a credit or debit card being presented from a customer to an online merchant. The system includes a peripheral device having a p…

Show why similar

Feedback

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👍 Relevant 👎 Not relevant

Optional: what did you expect to see instead?

Submit feedback

#8

Systems and methods for brokered authentication express seller links

Similarity
38%

US10068220 · 2018 · Visa International Service Association

G06Q Low confidence

Show abstract preview

Systems and methods for providing a transaction authorization from a funds facilitation system for a transaction requested by a user to a merchant system such that the user is not required to input any user authenticatio…

Show why similar

Feedback

Helps us learn what “relevant” means for students.

👍 **Relevant** 👎 **Not relevant**

Optional: what did you expect to see instead?

Submit feedback

#9

Device including encrypted data for expiration date and verification value creation

Similarity

38%

US10140598 · 2018 · Visa International Service Association

G06Q **Low confidence**

Show abstract preview

In order to make it more difficult to obtain numbers that can be used to conduct fraudulent transactions, a portion of a real account number is encrypted. The encrypted portion of the account number is used to generate a…

Show why similar

Feedback

Helps us learn what “relevant” means for students.

👍 **Relevant** 👎 **Not relevant**

Optional: what did you expect to see instead?

Submit feedback

#10

Transaction initiation determination system utilizing transaction data elements

Similarity

38%

US10176478 · 2019 · Visa International Service Association

G06Q | **Low confidence**

Show abstract preview

Systems and methods are described that allow for determining a transaction initiation mode used to conduct a transaction and applying a specific set of rules associated with the transaction initiation mode to the transac…

Show why similar

Feedback

Helps us learn what “relevant” means for students.

👍 **Relevant** 👎 **Not relevant**

Optional: what did you expect to see instead?

Submit feedback

Tip: If you’re getting “printer / messaging” type patents, add those words to **Exclude keywords**.